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## HEATHER MILLER

### Research Interests

Concurrent, distributed, eventually-consistent (edge computing), data-centric, and data-intensive (big data) programming, from the perspective of programming languages. More recently, my work has come to include *programming LLM systems*, or, focusing on how best to program *Compound AI Systems*. I work on both theoretical ideas & implementations. My goal is to reduce the burden of building distributed, and increasingly, AI-enabled systems.

### Education

*EPFL, Lausanne, Switzerland* 2009 – 2015  
Ph.D. in Computer Science  
Advisor: Martin Odersky 2011 – 2015  
  
*University of Miami, Coral Gables, FL* 2006 – 2009  
BSEE in Electrical Engineering, Audio Engineering, *with honors*, May 2009  
*Cooper Union for the Advancement of Science and Art, New York, NY* 2004 – 2006

### Employment

**Two Sigma Investments, New York City, NY, USA** 10/2022 –  
*Vice President, Research Scientist*  
Two Sigma Labs team, research interests: distributed programming, distributed systems, and programming LLMs.  
  
**Carnegie Mellon University, Pittsburgh, PA, USA** 8/2018 –  
*Assistant Professor*  
School of Computer Science, Software and Societal Systems Department  
Co-founder (with Ben L. Titzer) of the [Web Assembly Research Center](#)  
  
**Northeastern University, Boston, MA, USA** 9/2016 – 7/2018  
*Assistant Clinical Professor*  
College of Computer and Information Science  
  
**Scala Center, EPFL, Lausanne, Switzerland** 10/2015 – 7/2018  
*Executive Director, Research Scientist*  
Founded a new not-for-profit center dedicated to research, open source development, and education surrounding the Scala programming language.  
  
**Databricks, Berkeley, CA, USA** 8/2014 – 11/2014  
*Research Intern*  
Supervisor: Matei Zaharia  
Integrated Scala Pickling, our framework for fast, boilerplate-free, extensible serialization focused on distributed programming (OOPSLA'13), into Spark. Developed generalization of Spark/MapReduce programming model. (JFP'18).

### Teaching Experience (Classroom)

|                                                                                                                            |                                                                |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Co-Instructor,</b><br>15-440/15-640: Distributed Systems                                                                | <i>Fall 2020, 2022, 2023, 2024</i><br><i>Carnegie Mellon</i>   |
| <b>Instructor, Designer,</b><br>17-400/17-700: Data Science and Machine Learning at Scale                                  | <i>Fall 2020, Spring 2021</i><br><i>Carnegie Mellon</i>        |
| <b>Co-Instructor,</b><br>10-405/10-605: Machine Learning with Large Datasets                                               | <i>Spring 2020</i><br><i>Carnegie Mellon</i>                   |
| <b>Co-Instructor,</b><br>17-356: Software Engineering for Startups                                                         | <i>Spring 2019 &amp; Spring 2020</i><br><i>Carnegie Mellon</i> |
| <b>Instructor, Designer,</b><br>CS4240: Large-Scale Parallel Data Processing                                               | <i>Spring 2018</i><br><i>Northeastern</i>                      |
| <b>Instructor, Designer,</b><br>CS7680: Programming Models for Distributed Computation                                     | <i>Fall 2016</i><br><i>Northeastern</i>                        |
| <b>Co-Instructor, Co-Designer, (with Viktor Kunčak &amp; Martin Odersky)</b><br>CS 206: Parallelism & Concurrency          | <i>Spring 2016</i><br><i>EPFL</i>                              |
| <b>Co-Instructor, Co-Designer, (with Viktor Kunčak &amp; Martin Odersky)</b><br>CS 212: Reactive Programming & Parallelism | <i>Spring 2015</i><br><i>EPFL</i>                              |
| <b>(Lead) Teaching Assistant,</b><br>CS 201: Functional Programming                                                        | <i>Fall 2011-2014</i><br><i>EPFL</i>                           |

### Teaching Experience (MOOCs)

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| <b>Instructor, Designer, Big Data Analysis with Scala and Spark</b><br>Popular Coursera MOOC on big data analysis using Spark.<br><ul style="list-style-type: none"> <li>Designed lectures and produced lecture videos. Designed exercises and developed cloud-hosted automated graders.</li> <li>Between March-November 2017, over 120,000 registered learners.</li> </ul>                                                                                                 | <i>2017 –</i><br><i>Coursera</i>      |
| <b>Lead, Scala Specialization (mini-degree)</b><br>Responsible for EPFL's offering of a Scala <i>mini-degree</i> on Coursera.<br><ul style="list-style-type: none"> <li>Assembled offering of 4 Scala MOOCs, topped off with a capstone project. Taught and produced 1 course in the specialization and managed the development of the remaining 3 courses and the project.</li> </ul>                                                                                      | <i>2015 –</i><br><i>Coursera</i>      |
| <b>Lead, Functional Programming Principles in Scala</b><br>Popular Coursera MOOC on functional programming in Scala.<br><ul style="list-style-type: none"> <li>Lead teaching staff member, organized a team of graduate students, managed content production, designed course exercises with cloud-hosted grading, production of lecture videos, etc.</li> <li>&gt;400,000 learners across iterations &amp; largest completion rate for a course its size (~19%)</li> </ul> | <i>2012 – 2014</i><br><i>Coursera</i> |

**Book****Distributed Programming**

Heather Miller, Nat Dempkowski, James Larisch,  
Christopher Meiklejohn, and Philipp Haller

A textbook about the building blocks we use to build distributed systems. These range from the small, RPC, futures, actors, to the large; systems built up of these components like MapReduce and Spark. We explore issues and concerns central to distributed systems like consistency, availability, and fault tolerance, from the lens of the programming models and frameworks that the programmer uses to build these systems.

*Source (draft)*

**Publications:  
Recent Popular  
Media**
**The Shift from Models to Compound AI Systems**

Matei Zaharia, Omar Khattab, Lingjiao Chen, Jared Quincy Davis,  
Heather Miller, Chris Potts, James Zou, Michael Carbin,  
Jonathan Frankle, Naveen Rao, Ali Ghodsi

*Berkeley Artificial Intelligence Research (BAIR) Blog, February 18, 2024*

*Berkeley AI Blog  
(Feb 2024)*

**A Guide to Large Language Model Abstractions**

Peter Yong Zhong, Haoze He, Omar Khattab, Christopher Potts,  
Matei Zaharia, Heather Miller

*Two Sigma Insights, corporate blog, January 16, 2024*

*Two Sigma Blog  
(Jan 2024)*

**Publications:  
In Preparation**
**RTBAS: Defending LLM Agents Against Prompt Injection and Privacy Leakage**

Peter Yong Zhong, Siyuan Chen, Ruiqi Wang, McKenna McCall,  
Ben L Titzer, Heather Miller

*arXiv preprint arXiv:2502.08966*

*arXiv 2025*

**Publications:  
Journals**
**A Reduction Semantics for Direct-Style Asynchronous Observables**

Philipp Haller, Heather Miller

*Journal of Logical and Algebraic Methods in Programming, Volume 105, p75-111.*

*JLAMP 2019*

**A Programming Model and Foundation for Lineage-Based Distributed Computation**

Heather Miller, Philipp Haller, Normen Müller

*Journal of Functional Programming, Volume 28, e7.*

*Special Issue: Programming Languages for Big Data*

*JFP 2018*

**Publications:  
Conferences**
**ViBench: A Benchmark on Vibe Coding**

Peter Zhong, Pashootan Vaezipoor, Fuyang Cui, Vaibhav Kumar,  
James Austin, Azin Asgarian, Toby Ho, Paul Inder, Imen Kedir, Zhen Li,  
Nicholas Ondo, Asna Shafiq, Ibrahim Sheikh, Edouard Sioufi, Setareh Soltanieh,  
Ben Wilde, Jacky Zhao, Ryan Carelli, Heather Miller, Michele Catasta

*ACM Conference on AI and Agentic Systems*

*CAIS 2026*

- Debugging WebAssembly? Put some Whamm on it!** OOPSLA 2025  
Elizabeth Gilbert, Matthew Schneider, Zixi An, Suhas Thalanki,  
Wavid Bowman, Alexander Bai, Ben L. Titzer, Heather Miller  
*ACM SIGPLAN Conference on Object Oriented Programming, Systems,  
Languages and Applications*
- SMT: Fine-Tuning Large Language Models with Sparse Matrices** ICLR 2025  
Haoze He, Juncheng Li, Xuan Jiang, Heather Miller  
*International Conference on Learning Representations*
- DSPy: Compiling Declarative Language Model Calls  
into State-of-the-Art Pipelines** ICLR 2024 *spotlight*  
Omar Khattab, Arnav Singhvi, Paridhi Maheshwari, Zhiyuan Zhang,  
Keshav Santhanam, Sri Vardhamanan A, Saiful Haq, Ashutosh Sharma,  
Thomas T. Joshi, Hanna Moazam, Heather Miller, Matei Zaharia, Christopher Potts  
*International Conference on Learning Representations*
- Flexible Non-intrusive Dynamic Instrumentation for WebAssembly** ASPLOS 2024  
Ben L. Titzer, Elizabeth Gilbert, Bradley Wei Jie Teo, Yash Anand,  
Kazuyuki Takayama, Heather Miller  
*ACM International Conference on Architectural Support for  
Programming Languages and Operating Systems*
- Can My Microservice Tolerate an Unreliable Database?  
Resilience Testing with Fault Injection and Visualization** ICSE 2024 *Demo*  
Michael Assad, Christopher Meiklejohn, Heather Miller, Stephan Krusche  
*IEEE/ACM 46th International Conference on Software Engineering*
- Method overloading the circuit** SoCC 2022  
Christopher Meiklejohn, Lydia Stark, Cesare Celozzi, Matt Ranney, Heather Miller  
*ACM Symposium on Cloud Computing*
- Service-Level Fault Injection Testing** SoCC 2021  
Christopher Meiklejohn, Andrea Estrada, Yiwen Song, Rohan Padhye,  
Matt Ranney, Heather Miller  
*ACM Symposium on Cloud Computing*
- Composing and Decomposing Op-Based CRDTs with  
Semidirect Products** ICFP 2020  
Matthew Weidner, Christopher Meiklejohn, Heather Miller  
*ACM SIGPLAN International Conference on Functional Programming*
- Heard it Through the Gitvine: An Empirical Study of Tool  
Diffusion Across the npm Ecosystem** FSE 2020  
Hemank Lamba, Asher Trockman, Daniel Armanios, Christian Kästner,  
Heather Miller, Bogdan Vasilescu  
*ACM Symposium on the Foundations of Software Engineering*
- Partisan: Scaling the Distributed Actor Runtime** USENIX ATC 2019  
Christopher Meiklejohn, Heather Miller, Peter Alvaro  
*USENIX Annual Technical Conference*

**Scala Implicits are Everywhere: A Large-Scale Study of the Use of Implicits in the Wild** OOPSLA 2019

Filip Křikava, Heather Miller, Jan Vitek  
*ACM SIGPLAN Conference on Object Oriented Programming, Systems, Languages and Applications*

**Simplicity: Foundations and Applications of Implicit Function Types** POPL 2018

Martin Odersky, Olivier Blanvillain, Fengyun Liu, Aggelos Biboudis  
 Heather Miller, Sandro Stucki  
*ACM SIGPLAN Symposium on Principles of Programming Languages*

**Function Passing: A Model for Typed, Distributed Functional Programming** SPLASH 2016

Heather Miller, Philipp Haller, Normen Müller, Joceyln Boullier  
*ACM SIGPLAN International Symposium on New Ideas, New Paradigms, and Reflections on Programming & Software*

**Spores: A Type-Based Foundation for Closures in the Age of Concurrency and Distribution** ECOOP 2014

Heather Miller, Philipp Haller, Martin Odersky  
*European Conference on Object Oriented Programming*

**Functional Programming For All! Scaling a MOOC for Students And Professionals Alike** ICSE 2014

Heather Miller, Philipp Haller, Lukas Rytz, Martin Odersky  
*ACM SIGSOFT International Conference on Software Engineering*

**Instant Pickles: Generating Object-Oriented Pickler Combinators for Fast and Extensible Serialization** OOPSLA 2013

Heather Miller, Philipp Haller, Eugene Burmako, Martin Odersky  
*ACM SIGPLAN Conference on Object Oriented Programming, Systems, Languages and Applications*

**Publications:  
Workshops**

**For-Each Operations in Collaborative Apps** PaPoC 2023

Matthew Weidner, Ria Pradeep, Benito Geordie, Heather Miller  
*Workshop on Principles and Practice of Consistency for Distributed Data*

**Programmer Experience When Using CRDTs to Build Collaborative Webapps: Initial Insights** PLATEAU 2023

Yicheng Zhang, Matthew Weidner, Heather Miller  
*Workshop on the Intersection of Human Computer Interaction and Programming Languages*

**Checking-in on Network Functions** ANRW 2019

Zeeshan Lakhani, Heather Miller  
*ACM/IRTF Applied Networking Research Workshop*

**Towards a Solution to the Red Wedding Problem** USENIX HotEdge 2018

Christopher Meiklejohn, Heather Miller, Zeeshan Lakhani  
*USENIX Workshop on Hot Topics in Edge Computing*

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
|                          | <b>Distributed Programming via Safe Closure Passing</b><br>Philipp Haller, Heather Miller<br><i>Programming Language Approaches to Communication and Concurrency Centric Systems</i>                                                                                                                                                                                                                                                   | PLACES 2015   |
|                          | <b>RAY: Integrating Rx and Async for Direct-Style Reactive Streams</b><br>Philipp Haller, Heather Miller<br><i>ACM SPLASH Workshop on Reactivity, Events and Modularity</i>                                                                                                                                                                                                                                                            | REM 2013      |
|                          | <b>FlowPools: A Lock-Free Deterministic Concurrent Dataflow Abstraction</b><br>Aleksandar Prokopec, Heather Miller, Tobias Schlatter, Philipp Haller, Martin Odersky<br><i>International Workshop on Languages and Compilers for Parallel Computing</i><br>Invited to Revised Selected Papers on the 25th International Workshop on Languages and Compilers for Parallel Computing, Lecture Notes in Computer Science, Vol. 7760, 2013 | LCPC 2012     |
|                          | <b>Tools and Frameworks for Big Learning in Scala: Leveraging the Language for High Productivity and Performance</b><br>Heather Miller, Philipp Haller, Martin Odersky<br><i>NIPS Workshop on Parallel and Large-Scale Machine Learning</i>                                                                                                                                                                                            | BigLearn 2011 |
|                          | <b>Parallelizing Machine Learning – Functionally: A Framework and Abstractions for Parallel Graph Processing</b><br>Philipp Haller, Heather Miller<br><i>Scala Workshop</i>                                                                                                                                                                                                                                                            | Scala 2011    |
| Selected<br>Tech Reports | <b>The Function Passing Model: Types, Proofs, and Semantics</b><br>Philipp Haller, Normen Müller, Heather Miller                                                                                                                                                                                                                                                                                                                       | May 2016      |
|                          | <b>Specialising Parsers for Queries</b><br>Manohar Jonnalagedda, Jorge Vicente Cantero, Heather Miller, Martin Odersky                                                                                                                                                                                                                                                                                                                 | April 2016    |
|                          | <b>Improving Human-Compiler Interaction Through Customizable Type Feedback</b><br>Hubert Plociniczak, Heather Miller, Martin Odersky                                                                                                                                                                                                                                                                                                   | December 2014 |
|                          | <b>Self-Assembly: Lightweight Language Extension and Datatype Generic Programming, All-in-One!</b><br>Heather Miller, Philipp Haller, Bruno C. d. S. Oliveira                                                                                                                                                                                                                                                                          | August 2014   |
|                          | <b>Spores, Formally</b><br>Heather Miller, Philipp Haller                                                                                                                                                                                                                                                                                                                                                                              | December 2013 |
|                          | <b>FlowPools: A Lock-Free Deterministic Concurrent Dataflow Abstraction – Proofs</b><br>Aleksandar Prokopec, Heather Miller, Philipp Haller                                                                                                                                                                                                                                                                                            | June 2012     |
|                          | <b>General Chair and/or Program Chair:</b><br><i>Inaugural ACM Conference on AI and Agent Systems (ACM CAIS)</i><br><i>Compound AI Systems Workshop (Compound AI Systems)</i>                                                                                                                                                                                                                                                          | 2026<br>2024  |
| External<br>Service      |                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |

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|----------------------------------------------------------------------------------------|------------------------------|
| <i>ICSE Software Engineering in Practice (ICSE SEIP)</i>                               | 2022                         |
| <i>Curry On (Curry On)</i>                                                             | 2015, 2016, 2017, 2018, 2019 |
| <i>Workshop on Principles and Practice of Consistency for Distributed Data (PaPoC)</i> | 2019                         |
| <i>Trends in Functional Programming in Education (TFPIE)</i>                           | 2018                         |
| <i>Scala Symposium (Scala)</i>                                                         | 2013, 2014, 2017             |
| <i>Programming Models &amp; Languages for Distributed Computation (PMLDC)</i>          | 2016, 2017                   |

**Organizing Committee Member:**

|                                                                                    |             |
|------------------------------------------------------------------------------------|-------------|
| <i>Object-Oriented Programming, Systems, Languages &amp; Applications (OOPSLA)</i> | 2018        |
| <i>European Conference on Object-Oriented Programming (ECOOP)</i>                  | 2015 – 2019 |

**Program Committee Invitations Refused or Committees Dropped Out Of:** EuroSys'26, USENIX ATC ERC '25, Onward Papers'25, PLDI'25, SoCC'24, PaPoC'24, Onward Papers'24, SoCC'23, OOPSLA'23, Onward Essays'22

**Program Committee Member:**

|                                                                                                                   |      |
|-------------------------------------------------------------------------------------------------------------------|------|
| <i>European Conference on Object-Oriented Programming (ECOOP)</i>                                                 | 2024 |
| <i>International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)</i> | 2023 |
| <i>ACM Symposium on Cloud Computing Engineering (SoCC)</i>                                                        | 2022 |
| <i>International Conference on Software Engineering (ICSE)</i>                                                    | 2021 |
| <i>USENIX Workshop on Hot Topics in Cloud Computing (USENIX HotCloud)</i>                                         | 2020 |
| <i>USENIX Workshop on Hot Topics in Edge Computing (USENIX HotEdge)</i>                                           | 2020 |
| <i>Workshop on Principles and Practice of Consistency for Distributed Data (PaPoC)</i>                            | 2020 |
| <i>Object-Oriented Programming, Systems, Languages &amp; Applications (OOPSLA)</i>                                | 2019 |
| <i>European Conference on Object-Oriented Programming (ECOOP)</i>                                                 | 2019 |
| <i>Symposium on Principles of Programming Languages (POPL)</i>                                                    | 2019 |
| <i>International Conference on Functional Programming (ICFP)</i>                                                  | 2018 |
| <i>Off the Beaten Track (OBT)</i>                                                                                 | 2018 |
| <i>Object-Oriented Programming, Systems, Languages &amp; Applications (OOPSLA)</i>                                | 2017 |
| <i>Scala Symposium (Scala)</i>                                                                                    | 2016 |
| <i>Symposium on Trends in Functional Programming (TFP)</i>                                                        | 2016 |
| <i>Software Language Engineering (SLE)</i>                                                                        | 2016 |
| <i>Symposium on Applied Computing (SAC)</i>                                                                       | 2016 |
| <i>Programming Language Evolution (PLE)</i>                                                                       | 2015 |
| <i>Domain-Specific Language Design and Implementation (DSLDI)</i>                                                 | 2015 |

**External Review Committee Member:**

PLDI 2020, PLDI 2018, ECOOP 2016, ECOOP 2013, Scala 2013

**Artifact Evaluation Committee:** POPL 2015

**Diversity & Outreach**

**Confluence Talks Co-Creator/Organizer**

Co-created a new talk series at CMU intent on building a bridge between Pittsburgh's local tech scene and industry-relevant research at CMU.

**ScalaBridge Organizer**

Organizer of free full-day workshops on the weekends aimed at teaching women and underrepresented minorities in computing how to think computationally and how to program in Scala.

*ScalaBridge Chapters: Basel (CH), Zürich (CH), Copenhagen (DK), Boston (US).*

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|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Open Source    | Scala Programming Language, <i>member of the Scala team</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2011 –                                    |
|                | <ul style="list-style-type: none"> <li>• <a href="#">Scala Spores</a> (Scala Improvement Proposal SIP-21), <i>project lead</i><br/>novel type-based abstraction for using closures safely in concurrent and distributed environments</li> <li>• <a href="#">Scala Pickling</a>, <i>project lead</i><br/>novel framework for fast, boilerplate-free, extensible serialization. Adopted by sbt, the most widely-used build tool for Scala. Popular open-source project on GitHub with &gt;820 stars &amp; dozens of contributors</li> <li>• <a href="#">Scala Futures &amp; Promises</a> (Scala Improvement Proposal SIP-14), <i>team member</i><br/>unified non-blocking concurrency substrate for Scala, Akka, Play, and others</li> <li>• <a href="#">Scala Documentation</a>, <i>creator, writer, lead maintainer</i><br/>a central website for community-driven documentation for the Scala programming language and core libraries</li> <li>• <a href="#">Scaladoc</a>, <i>co-maintainer</i><br/>documentation tool for Scala's official API documentation</li> </ul> |                                           |
|                | <a href="#">Dahl-Nygaard Junior Prize</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2023                                      |
|                | ACM SIGPLAN Programming Languages Software Award (for Scala)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2019                                      |
|                | US National Science Foundation Graduate Research Fellowship                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2011 – 2014                               |
|                | EPFL Outstanding Teaching Award                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2012                                      |
| Honors         | EPFL Computer Science Fellowship                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2009 – 2010                               |
|                | Most Outstanding Audio Engineering Student, University of Miami                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2009                                      |
|                | Most Outstanding Eta Kappa Nu Student, University of Miami                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2009                                      |
|                | Information Technology Scholarship, University of Miami                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2006 – 2009                               |
|                | John Farina Family Scholarship, University of Miami                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2006 – 2009                               |
|                | Eta Kappa Nu                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2008                                      |
|                | Tau Beta Pi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2008                                      |
|                | SMART US Department of Defense Scholarship Alternate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2007                                      |
|                | Cooper Union Full Tuition Scholarship                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2004 – 2006                               |
|                | <b>Open Source Numbers Everybody Should Know</b> <i>Open Source Summit North America</i><br>Austin TX, USA (held virtually). June 29, 2020                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | (keynote)                                 |
|                | <b>Open Source Numbers Everybody Should Know</b><br>Berlin, Germany. February 28, 2020                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | BOBKonf 2020<br>(keynote)                 |
| Selected Talks | <b>The Times They Are a-Changin': A Data-Driven Portrait of New Trends in How We Build Software, Open Source, &amp; What Even is Entry-Level Now</b><br>Oakland, CA, USA. November 14, 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Scale By the Bay</i> 2019<br>(keynote) |
|                | <b>Scala Implicits are Everywhere: A Large-Scale Study of the Use</b><br>Athens, Greece. October 24, 2019                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | OOPSLA 2019                               |
|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                           |



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| <b>We're Building On Hollowed Foundations: Worrying Trends in Open Source and What We Can Actually Do About It</b><br>Genoa, Italy. April 4, 2019                        | <i>Programming 2019<br/>(keynote)</i>    |
| <b>Towards Language Support for Distributed Systems</b><br>London, UK. November 9, 2018                                                                                  | <i>Code Mesh 2018<br/>(invited)</i>      |
| <b>What Happened to Distributed Programming Languages?</b><br>Boston, MA, USA. November 6, 2018                                                                          | <i>SPLASH-I 2018<br/>(invited)</i>       |
| <b>Towards Language Support for Distributed Systems</b><br>St. Louis, MO, USA. September 27, 2018                                                                        | <i>Strange Loop 2018</i>                 |
| <b>I'm a Young Assistant Professor: AMA. + Heather's Unsolicited Advice About Grad School</b><br>St. Louis, MO, USA. September 23, 2018                                  | <i>PLMW 2018<br/>(invited)</i>           |
| <b>We're Building On Hollowed Foundations: Worrying Trends in Open Source and What You Can Actually Do About It</b><br>Krakow, Poland. February 22, 2018                 | <i>Lambda Days 2018<br/>(keynote)</i>    |
| <b>The Dramatic Consequences of the Open Source Revolution: Unrecognized Challenges &amp; Some Modest Attempts at Solutions in Scala</b><br>Paris, France. April 7, 2017 | <i>Devoxx 2017<br/>(invited)</i>         |
| <b>The Dramatic Consequences of the Open Source Revolution &amp; How the Scala Center Hopes to Help</b><br>London, UK. December 9, 2016                                  | <i>Scala Exchange 2016<br/>(keynote)</i> |
| <b>Function Passing: A Model for Typed, Distributed Functional Programming</b><br>Amsterdam, The Netherlands. November 2, 2016                                           | <i>SPLASH 2016</i>                       |
| <b>Introducing the Scala Center</b><br>New York, NY, US. May 10, 2016 & Berlin, Germany. June 16, 2016<br>(total ~1700 attendees)                                        | <i>Scala Days 2016<br/>(keynote)</i>     |
| <b>Function Passing Style: Typed, Distributed Functional Programming</b><br>St. Louis, MO, USA. September 19, 2014                                                       | <i>Strange Loop 2014</i>                 |
| <b>Spores: A Type-Based Foundation for Closures in the Age of Concurrency and Distribution</b><br>Uppsala, Sweden. August 1, 2014                                        | <i>ECOOP 2014</i>                        |
| <b>Functional Programming For All! Scaling a MOOC for Students and Professionals Alike</b><br>Hyderabad, India. June 4, 2014                                             | <i>ICSE 2014</i>                         |
| <b>Academese to English: Scala's Type System, Dependent Types</b>                                                                                                        | <i>NEScala 2014</i>                      |

**and What It Means To You**

New York, NY, USA. March 1, 2014

**Instant Pickles: Generating Object-Oriented Pickler Combinators for Fast and Extensible Serialization**

OOPSLA 2013

Indianapolis, IN, USA. October 30, 2013

**PL Abstractions for Distributed Programming: Pickle Your Spores!**Indiana University (*invited*)

Bloomington, IN, USA. October 25, 2013

**Spores: Distributable Functions in Scala**

Strange Loop 2013

St. Louis, MO, USA. September 19, 2013

**Open Issues in Dataflow Programming**LaME 2013 (*invited*)

Montpellier, France. July 1, 2013

**Scala as a Research Tool**

ECOOP 2013 Tutorial

Montpellier, France. July 1, 2013

**On Pickles & Spores: Improving Scala's Support for Distributed Programming**

ScalaDays 2013

New York, NY, USA. June 12, 2013

**Futures & Promises in Scala 2.10**PhillyETE 2013 (*invited*)

Philadelphia, PA, USA. April 2, 2013

*I am also a frequent speaker in industry, at industrial conferences, developer “meet-ups”, and everything in between. Some such events include:*

Scala Italy (9/2018, Florence, Italy), LxScala (6/2018, Lisbon, Portugal), Open Source Summit (12/2017, Paris, France), Scala World (9/2017, Lake District, UK), LxScala (5/2017, Lisbon, Portugal), Lambda Days (2/2017, Krakow, Poland), PhillyETE (4/2016, Philadelphia, USA), Code Mesh (11/2015, London, UK), Scalar (4/2015, Warsaw, Poland), f(by) (11/2014, Minsk, Belarus), SF Scala (11/2014, SF, USA), Scalapeño (9/2014, Tel Aviv, Israel), SoundCloud TechTalks (7/2014, Berlin, Germany), Scala Days (6/2014, Berlin, Germany), NEScala (3/2014, NYC, USA), amongst others.

**External Activities**

Scalawags Monthly Podcast, co-host

2014 – 2016

**Students Supervised**

**Siyuan Chen** (co-advised with Phil Gibbons and Ben L. Titzer)  
PhD thesis

2023 –  
Carnegie Mellon

**Peter Yong Zhong**  
PhD thesis

2023 –  
Carnegie Mellon

**Haoze Hector He**  
PhD thesis

2023 –  
Carnegie Mellon

|                                                                                                                                 |                                |
|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>Elizabeth Gilbert</b> (co-advised with Ben L. Titzer)<br>PhD thesis                                                          | 2022 –<br>Carnegie Mellon      |
| <b>Matthew Weidner</b><br><i>Increasing the Flexibility of Collaborative Data Structures</i><br>PhD thesis                      | 2019 –<br>Carnegie Mellon      |
| <b>Dr. Christopher Meiklejohn</b><br><i>Resilient Microservice Applications, By Design, and without the Chaos</i><br>PhD thesis | 2018 – 2024<br>Carnegie Mellon |
| <b>Joeyln Boullier</b> , <i>Evaluating the Efficacy of the Function Passing Model</i><br>M.Sc. thesis                           | 2/2016 – 8/2016<br>EPFL        |
| <b>Jorge Vicente Cantero</b> , <i>Implementing the Function Passing Model</i><br>B.Sc. thesis                                   | 2/2016 – 6/2016<br>EPFL        |
| <b>Thaddée Yann Tyl</b> , <i>Learning Scala Style</i><br>M.Sc. thesis                                                           | 2/2013 – 6/2013<br>EPFL        |

## References

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